RevTrace: A Transactional Runtime for Feedback-State Integrity in Revision-Generative ASR Serving

Bin Zhou, Jiyuan Miao, and Xi Cao

***Abstract*—Revision-generative automatic speech recognition (ASR) pipelines feed model outputs back into later execution state. Under batched accelerator execution, a local numerical difference can therefore become persistent state contamination. We formalize this failure mode and introduce Dual-Granularity Feedback Transactions (DGFT), a runtime protocol that separates low-latency proposal and display from canonical state-write authority. Each accepted projection is bound to a certificate, parent version, input frontier, and runtime configuration; same-granularity certification is the $\rho = 1$ case. RevTrace implements DGFT as a two-plane parallel runtime with independently replicated proposers and authorities. Across Qwen3-ASR and Whisper on two RTX 3090 GPUs, coarse authority improves matched-workload throughput by $1.39\times$/$1.28\times$ and reaches up to $6.88\times$ wall-clock speedup over per-path all-singleton references, with no observed installation violation in 800 Qwen events and 1,048 Whisper commits. A 250,000-attempt protocol fuzzer rejects every mutation and replay. On eight V100 GPUs, balanced replication sustains $157.8\times$/$111.7\times$ audio real-time throughput at the largest measured concurrency satisfying the 2-s display service-level objective (SLO). These results establish state-write authorization as a distinct runtime contract for feedback-driven ASR serving.**

***Index Terms*—runtime systems, generative model serving, streaming ASR, state certification, transactional consistency**

I. Introduction

Accelerator-backed generative services usually treat a returned sequence as an endpoint. Revision-generative ASR does not: Whisper-style decoders condition on previously generated text [1], [2], and Qwen3-ASR re-prompts itself with rollback prefixes from earlier revisions [3]. A generated hypothesis therefore mutates the state of later execution. Yet throughput-oriented runtimes optimize request scheduling and output delivery without exposing an authorization boundary between model generation and this feedback write [4]–[6].

Batching improves accelerator utilization by sharing execution across sessions [4], [7]–[9], but it can change floating-point reduction order and hence a greedy token sequence relative to singleton execution [10]. A terminal service contains that difference within one response. In a revision-generative service, the differing output can become the parent of every later revision (Fig. 1). Controlled paired interventions establish that this is a feedback-channel failure, not merely display variation: 8 of 21 legal state pairs propagate when the feedback edge is present, none propagate when it is removed, and the measured revision amplification factor (RAF) reaches 153.

Existing mechanisms regulate different boundaries. LocalAgreement stabilizes display prefixes but does not authorize feedback parents [2]; speculative decoding verifies draft tokens against a target distribution [11], [12]; and LLM-42 provides verify–rollback for deterministic token generation, not revision-state installation [13]. Numerical reproducibility repairs execution rather than state provenance [10]. Optimistic concurrency control supplies a useful analogy [14], [15], but here the speculative write is a model-generated projection whose legality must be established by the serving runtime.

We accordingly treat each revision output as a speculative state write: $\mathrm{Display}(Y_t) \not\Rightarrow \mathrm{Authorize}(\Phi(Y_t; H_t))$. We introduce **Dual-Granularity Feedback Transactions** (DGFT), which decouples a fine-grained proposal/display path from a coarse canonical authority. Only a certificate bound to the parent version, input frontier, projection, and execution contract can install state; same-granularity certification is the $\rho = 1$ case. **RevTrace** implements DGFT as a two-plane parallel runtime with contract-scoped certification. The abstraction targets feedback-driven generation, while all empirical generality claimed here is explicitly limited to the evaluated revision-generative ASR pipelines.

This paper contributes:

1) a formal feedback-state contamination model and causal evidence that identifies the feedback edge as the propagation channel (Sections II and VI);
2) DGFT and the RevTrace runtime, with certificate-, version-, and frontier-bound commits, three state-history invariants, and a cost model for selecting authority granularity (Sections III–V); and
3) a contract-bounded evaluation on Qwen3-ASR 0.6B/1.7B and Whisper tiny/base/small.en, using two RTX 3090 GPUs and an independent $8\times$V100 platform. It covers matched-workload throughput, display and commit latency, energy, family-specific quality, protocol fuzzing, and the measured single-node capacity boundary (Section VI).

II. Model and Problem Formulation

A. Revision-generative state machines

Definition II.1 (Revision-generative serving round)**.** *Round t is (S_t, A_t, Y_t), where S_t is committed state, A_t is exogenous input*

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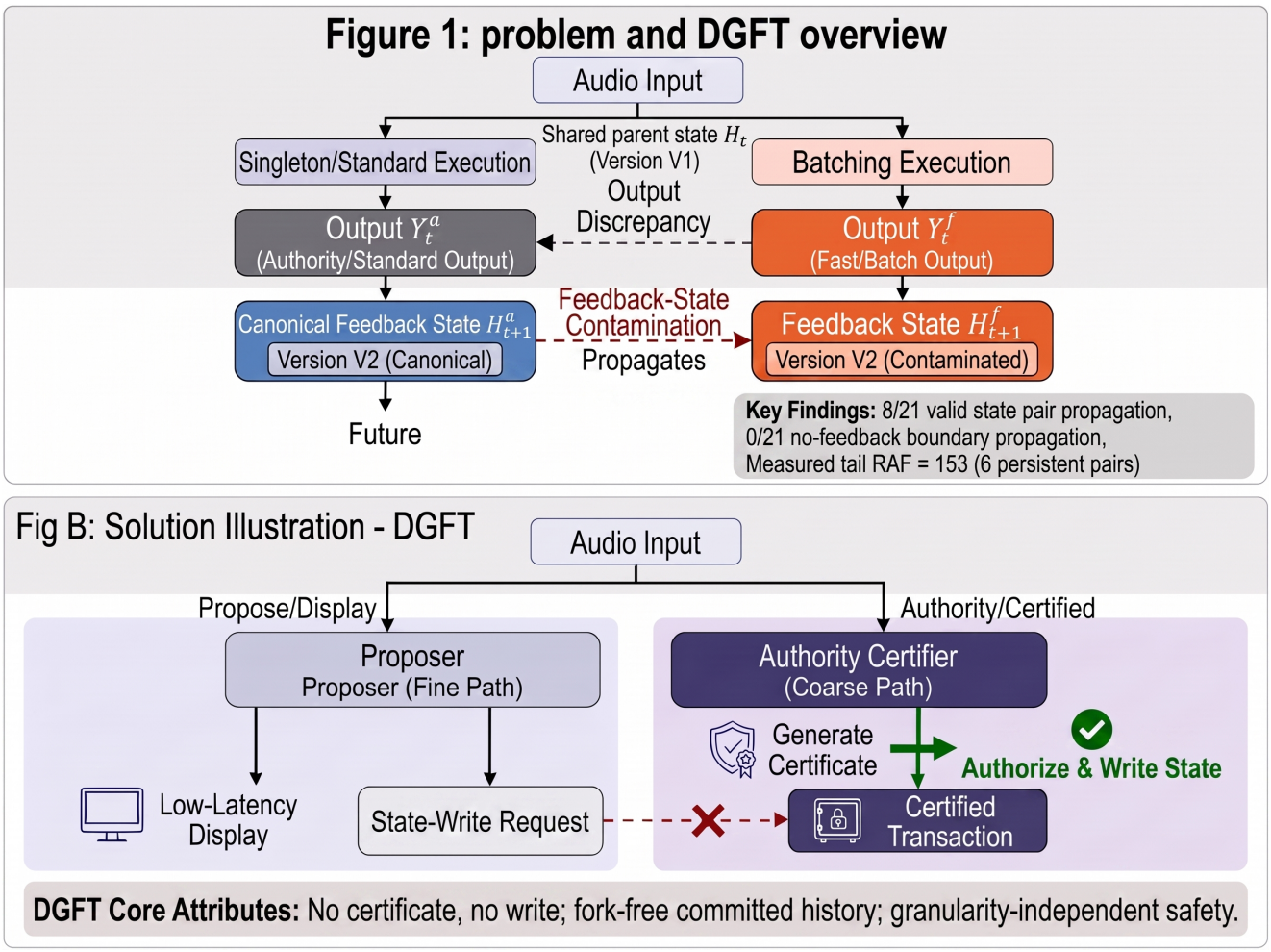


Fig. 1. State-entry contamination under ordinary batching (a) and DGFT (b). Provisional outputs may differ; only certificate-authorized projections may become parents of future state.

(the audio frontier), and Y_t is generated output. Its update is

$$S_{t+1} = \mathcal{T}(S_t, A_t, Y_t). \tag{1}$$

The system is revision-generative *when $\mathcal{T}$ depends on Y_t, so round t affects state read by round $t{+}1$.*

Definition II.2 (Feedback sub-state and projection)**.** *Write $S_t = (S_t^x, H_t)$, where H_t is generated feedback and S_t^x contains exogenous/runtime conditions held fixed in a causal pair. The adapter projection Φ writes*

$$H_{t+1} = \Phi(Y_t; H_t), \tag{2}$$

and the full update factors as

$$S_{t+1} = \mathcal{G}(S_t^x, H_{t+1}, A_t). \tag{3}$$

For Qwen3-ASR, H_t is the rollback prefix [3]; for Whisper with previous-text conditioning, it is the prompt [1].

The *feedback edge* is the dependency $Y_t \to H_{t+1}$ in Eq. (2); removing it makes the next feedback state independent of Y_t while holding future inputs and runtime assignments fixed. Ordinary batching instead installs

$$H_{t+1} = \Phi(Y_t^{\text{batch}}; H_t), \tag{4}$$

without a separate state-write decision.

B. Display output versus state projection

Display-only suffixes, hold-back regions, and provisional hypotheses may be visible while excluded from Φ by the adapter. DGFT therefore enforces the permission boundary

$$\boxed{\text{Display}(Y_t) \;\not\Rightarrow\; \text{Authorize}(\Phi(Y_t; H_t)).} \tag{5}$$

C. State-entry divergence and contamination

For the same round, let Y_t^f and Y_t^a denote fine (batched) proposal and canonical authority outputs.

Definition II.3 (Output divergence)**.** *$\varepsilon_t = d_Y(Y_t^f, Y_t^a)$, where d_Y is the contract's output-space distance.*

Definition II.4 (State-entry divergence)**.** *For $P_t^f = \Phi(Y_t^f; H_t)$ and $P_t^a = \Phi(Y_t^a; H_t)$, $e_t = d_S(P_t^f, P_t^a)$ under the same mixed-unit metric.*

Definition II.5 (Contamination event)**.**

$$C_t \iff e_t > 0 \;\wedge\; \Phi(Y_t^f; H_t) \text{ is installed into } H_{t+1}.$$

Thus output difference alone is not contamination; it must acquire future-state write permission.

D. Feedback amplification

Let $D_t = d_H(H_t^f, H_t^a)$ be trajectory distance under identical future exogenous inputs and assignments.

Assumption II.1 (Local sensitivity)**.** *Within the evaluated neighborhood of the canonical trajectory, finite L_t, β_t satisfy*

$$D_{t+1} \le L_t D_t + \beta_t e_t. \tag{6}$$

The coefficients are unknown and the assumption makes no global Lipschitz claim; the decomposition identifies propagation channels only.

Remark II.2 (Causal prediction)*.* Removing the feedback edge sets $\beta_t = 0$ for the intervened transition, so an isolated state-entry perturbation cannot propagate through that edge when future inputs, assignments, and arithmetic realizations are equal. A fresh later divergence remains possible. Section VI tests this scoped prediction.

Definition II.6 (Revision amplification factor)**.** *For horizon K,*

$$\mathrm{RAF}_{t,K} = \frac{D_{t+K}}{e_t + \epsilon_0}$$

is a measured trajectory indicator, not a theoretical bound.

E. Design requirements

The model yields four requirements for certified feedback state:

- R1 (Authority separation): Display and state-write permissions are distinct (Eq. (5)).
- R2 (Parent ownership): A projection commits only to the parent version against which it was computed.
- R3 (Frontier monotonicity): Committed audio intervals advance without gaps or overlap, forming one history.
- R4 (Evidence non-authority): Statistical evidence may schedule verification but never authorize installation.

Section III realizes R1–R4 as protocol checks; Fig. 2 and Table I trace them through the runtime and baselines.

III. DGFT: Theory and Mechanism

A. Transactions, certificates, and commits

DGFT treats every projection written into the committed feedback sub-state H as a transaction; generated output never directly mutates H.

a) Trust model: The authority coordinator and versioned ledger are trusted; proposers and clients cannot mint or alter certificates. We assume collision-resistant digests, immutable certified audio bytes, and atomic certificate consumption with ledger compare-and-swap. A digest binds content but does not authenticate it independently; authenticity follows from protected authority issuance. We exclude Byzantine authorities and compromised ledgers. The protocol requires neither deterministic proposers nor synchronous GPU execution: the authority may use any decoding policy whose realized projection and complete runtime configuration, including its randomness policy, are bound by the certificate. Asynchronous completions remain safe because the parent-version compare-and-swap serializes installation. These assumptions certify provenance and write legality, not semantic correctness or repeatability of an independently rerun stochastic authority.

Definition III.1 (DGFT transaction)**.** *A transaction is*

$$\tau_t = (\mathrm{sid}, e, p, v_{\mathrm{parent}}, [a_s, a_e), Y_t, \Phi(Y_t; H_t), \mathrm{cert}_t),$$

where the fields identify the session, contract epoch, policy, parent version, covered audio frontier, output, state projection, and certificate. Proposal and authority granularities g_p and g_a define $\rho = g_a/g_p \ge 1$.

Definition III.2 (Certificate)**.** *For canonical projection $\Phi_j^a = \Phi(Y_j^a; H_{v_j})$, the authority issues the single-use record*

$$\mathrm{cert}_j = \mathrm{Bind}\big(\mathrm{sid}_j, e_j, p_j, v_j, [f_j, f_{j+1}), h(A_{[f_j, f_{j+1})}), h(\Phi_j^a), \chi_j\big),$$

where χ_j binds the checkpoint, tokenizer, decoding policy, projection adapter, precision, and authority-runtime version. Thus one certificate names exactly one canonical projection and commit context.

Definition III.3 (Commit validity)**.** *For*

$$\Gamma_j = (\mathrm{sid}_j, e_j, p_j, v_j, [f_j, f_{j+1}), \Phi_j^a, \mathrm{cert}_j),$$

the commit is valid iff: (i) its session, epoch, and policy match the live fence; (ii) v_j is the committed parent; (iii) its frontier is the next gap- and overlap-free interval; and (iv) the unconsumed certificate binds all presented fields and the recomputed projection digest. Only valid commits mutate H.

These checks realize R1–R4 (Section II-E): display creates no commit, parent and frontier are checked at installation, and risk evidence never issues a certificate. Table I contrasts this contract with weaker regimes. Binding χ_j is essential: a result from another checkpoint, tokenizer, precision, or decoder policy cannot be replayed as if it were produced by the live canonical contract.

Definition III.4 (Committed provenance ancestry)**.** *A projection is an ancestor of H_j iff it labels an accepted transition on the unique ledger path $H_0 \to \cdots \to H_j$; displayed or rejected projections are not ancestors.*

B. Theorem 1: uncertified-ancestry exclusion

Theorem III.1 (Uncertified-Ancestry Exclusion)**.** *If H changes only through valid commits, no fine projection unbound by a valid certificate can enter the ancestry of any future committed state.*

Proof sketch. Induct over accepted commits. H_0 has no generated ancestor. A valid next commit installs exactly the projection digest bound by its certificate; different content fails validation, while display does not mutate H. Therefore the invariant is preserved. The full proof, including the case in which proposal and authority projections coincide, is in Supplementary Section S2. □

The result depends on commit provenance, not margins, model determinism, precision agreement, or ASR architecture.

TABLE I
GUARANTEE SCOPE. ✓DENOTES A GUARANTEE WITHIN THE DECLARED CONTRACT; "DISPLAY ONLY" DOES NOT AUTHORIZE FEEDBACK-STATE INSTALLATION. THE FOUR EVIDENCE COLUMNS ARE NON-TRANSFERABLE: SG RISK UCB, AUTHORITY PREFLIGHT, TRANSACTION VALIDATION, AND FAMILY-SPECIFIC QUALITY ANSWER DIFFERENT QUESTIONS. MARGIN SIGNALS SCHEDULE VERIFICATION BUT AUTHORIZE NO WRITE.

Regime	Display/state-write authority split	Feedback ancestry	Canonical authority	Stale rejection	Statistical risk bound	Preflight invariance	Exhaustive tx validation	Semantic reference
Unprotected FP16 batching	×	×	×	×	×	×	×	×
Owner/version check only	×	×	×	✓	×	×	×	×
LocalAgreement-style confirmation	display only	×	×	×	×	×	×	×
Margin/confidence gate	×	×	×	×	×	×	×	×
SG-DGFT ($\rho = 1$)	✓	✓	✓	✓	✓	×	✓	✓
Coarse/Hybrid DGFT ($\rho > 1$)	✓	✓	✓	✓	×	✓	✓	✓

C. Theorem 2: fork-free committed-state linearity

Theorem III.2 (Fork-Free Committed-State Linearity)**.** *If validation and installation form one atomic compare-and-swap and each certificate is consumed at most once, parent-version and frontier checks make the accepted commits of a session a fork-free, gap-free, overlap-free linear history.*

Proof sketch. Atomic compare-and-swap prevents two commits from installing against the same parent. Parent checking rejects stale branches, single-use certificates reject replay, and frontier continuity orders the remaining intervals without gaps or overlaps. See Supplementary Section S2 for the stepwise argument. □

Remark III.3 (Scope of the concurrency claim). The ledger compare-and-swap serializes accepted commits. This is not full Herlihy–Wing linearizability of all client operations, which would require a real-time concurrent-object specification [16].

Corollary III.4 (Stale-descendant rejection)**.** *Any revision with* $v_{\text{parent}} \neq \text{version}(H_j)$ *fails the parent check and cannot mutate state.*

D. Theorem 3: granularity-independent safety

Theorem III.5 (Granularity-Independent Safety)**.** *For any* $\rho \geq 1$*, if each authority segment begins at the committed parent, covers the next contiguous frontier, produces a legal projection, and passes certificate and version checks, Theorems III.1 and III.2 hold.*

Proof sketch. Both proofs depend only on binding, validity, and atomic installation; changing ρ changes certificate coverage, not admissibility. □

Corollary III.6 (Same-granularity certification as $\rho = 1$)**.** *At* $\rho = 1$*, proposal and authority regions coincide: directional coverage becomes identity and fine replay becomes risk-triggered re-verification. SG is therefore a protocol reduction of DGFT.*

Remark III.7 (Canonical-path boundary). This reduction is protocol-level. Our SG contract uses FP16 singleton authority, whereas Coarse/Hybrid use FP32 batched authority; we never claim bitwise identity between these canonical paths.

E. Authority-cost decomposition

Let B denote concurrent sessions, q the SG verification fraction, r the Hybrid replay fraction, C_s singleton authority cost per fine region, $C_c(\rho, B)$ batched coarse cost, C_{commit} transaction overhead, and Q_s singleton queueing/fragmentation cost. Then

$$T^a_{\text{SG}} \approx qB\rho C_s + Q_s, \tag{7}$$

$$T^a_{\text{DGFT}} \approx C_c(\rho, B) + rB\rho C_s + C_{\text{commit}}, \tag{8}$$

and, with separate proposer and authority GPUs,

$$T_{\text{service}} \approx \max(T_p, T^a) + T_{\text{repair}}. \tag{9}$$

Proposition III.8 (Model-based operating condition)**.** *When authority is on the critical path and repair costs are comparable, Coarse-DGFT improves throughput over SG if*

$$C_c(\rho, B) + rB\rho C_s + C_{\text{commit}} < qB\rho C_s + Q_s.$$

The model predicts that even small q can create singleton tails, whereas a coarse authority may amortize invocation and queueing. Hybrid trades replay for quality, and model-dependent authority paths can reverse the ordering. Section VI-D tests these predictions; because the evaluated canonical paths differ (Remark III.7), results identify complete operating points rather than a precision-independent effect of ρ. The decomposition is therefore an explicit performance model, not part of the safety proof and not an assertion that coarse authority is always faster.

F. Operating-point trade-offs

Deployment selects SG, Coarse, or Hybrid to maximize an objective $J(\pi)$ subject to zero transaction violations and contract caps:

$$\max_{\pi} J(\pi) \quad \text{s.t.} \quad V(\pi) = 0,\ C(\pi) \leq B.$$

DGFT defines this safe envelope; it does not claim that one measured operating point dominates across workloads or models.

IV. THE REVTRACE RUNTIME

RevTrace realizes DGFT with family adapters, a two-plane coordinator, a versioned ledger, and contract-scoped certification (Fig. 2).

A. Adapters and two-plane coordination

Each model family supplies three interfaces: a projection adapter computing $\Phi(Y_t; H_t)$ (Qwen rollback prefix or Whisper previous-text prompt), an authority adapter executing its canonical path, and an evidence adapter that collects margins, directional coverage, and repetition signals. The

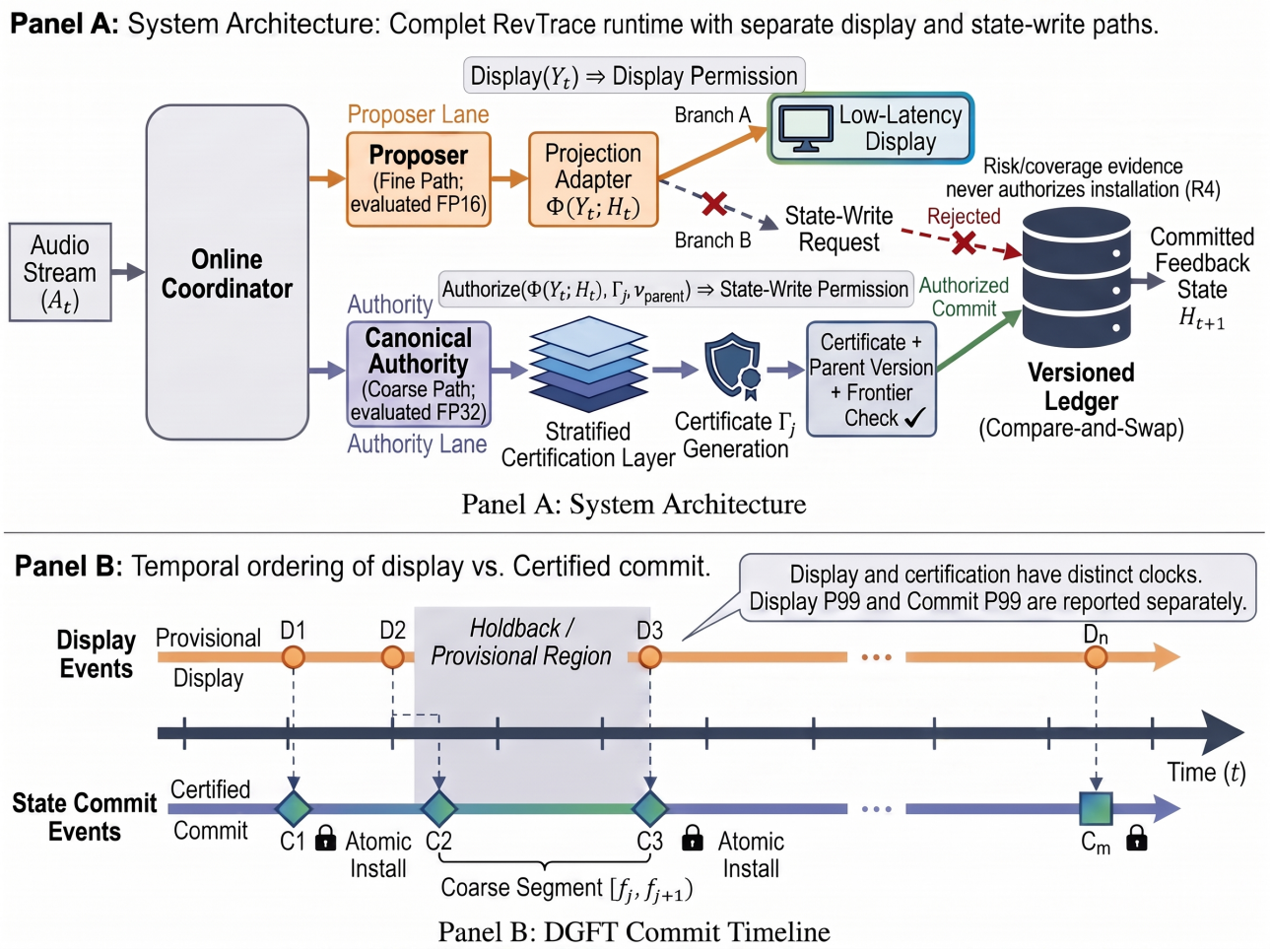


Fig. 2. RevTrace architecture and commit timeline. Proposal may drive low-latency display, but only a certificate bound to the current parent and frontier may install state. FP32 is the evaluated coarse authority, not a protocol requirement.

online coordinator runs proposer and authority on separate GPUs, invalidates stale descendants (Corollary III.4), closes terminal frontiers, and rebases displayed provisional state after superseding commits. RQ3 evaluates the same runtime through both family adapters. A slow authority may delay certified commit without blocking provisional display; the evaluation consequently reports display and commit clocks separately.

B. Certificate ledger

The compare-and-swap ledger stores the committed version and audio frontier. Before installation it matches each Γ_j against the session, contract epoch, policy, parent-version, frontier, and certificate-consumption fences. RQ2 exhaustively records these checks for every install attempt.

C. Coverage and selective replay

A coarse certificate records, in Latin or CJK units, which proposal regions it covers. Uncovered regions enter fine replay; Hybrid additionally replays risk-selected regions at fraction r. Coverage therefore controls work but does not grant write authority. RQ3 compares Coarse ($r = 0$) and Hybrid ($r > 0$) under a matched workload.

D. Stratified certification

The four non-transferable strata in Fig. 3 are: (i) randomized, anytime-valid UCB audits of SG numerical risk, scoped only to the SG adapter path [17], [18]; (ii) fail-closed, contract-scoped preflight tests of authority repeatability and batch-vs-singleton invariance; (iii) exhaustive online certificate, version, and frontier validation; and (iv) independent family-specific semantic references. Evidence may schedule verification but never authorize a state write (R4). The separation prevents a favorable statistical bound in one contract from being promoted into an installation certificate for another.

V. Implementation

RevTrace supports Qwen rollback-prefix and Whisper previous-text adapters with growing audio and holdback semantics [1], [3]. In the frozen implementation, SG uses FP16 singleton authority and Coarse/Hybrid use FP32 batched authority with a 4,096-token per-session KV-cache bound. The latter's 10.7–11.1 s one-time fail-closed preflight is reported separately and excluded from steady-state timing. FP32 is not a DGFT requirement: replacing it requires a new canonical contract and the same preflight and quality checks.

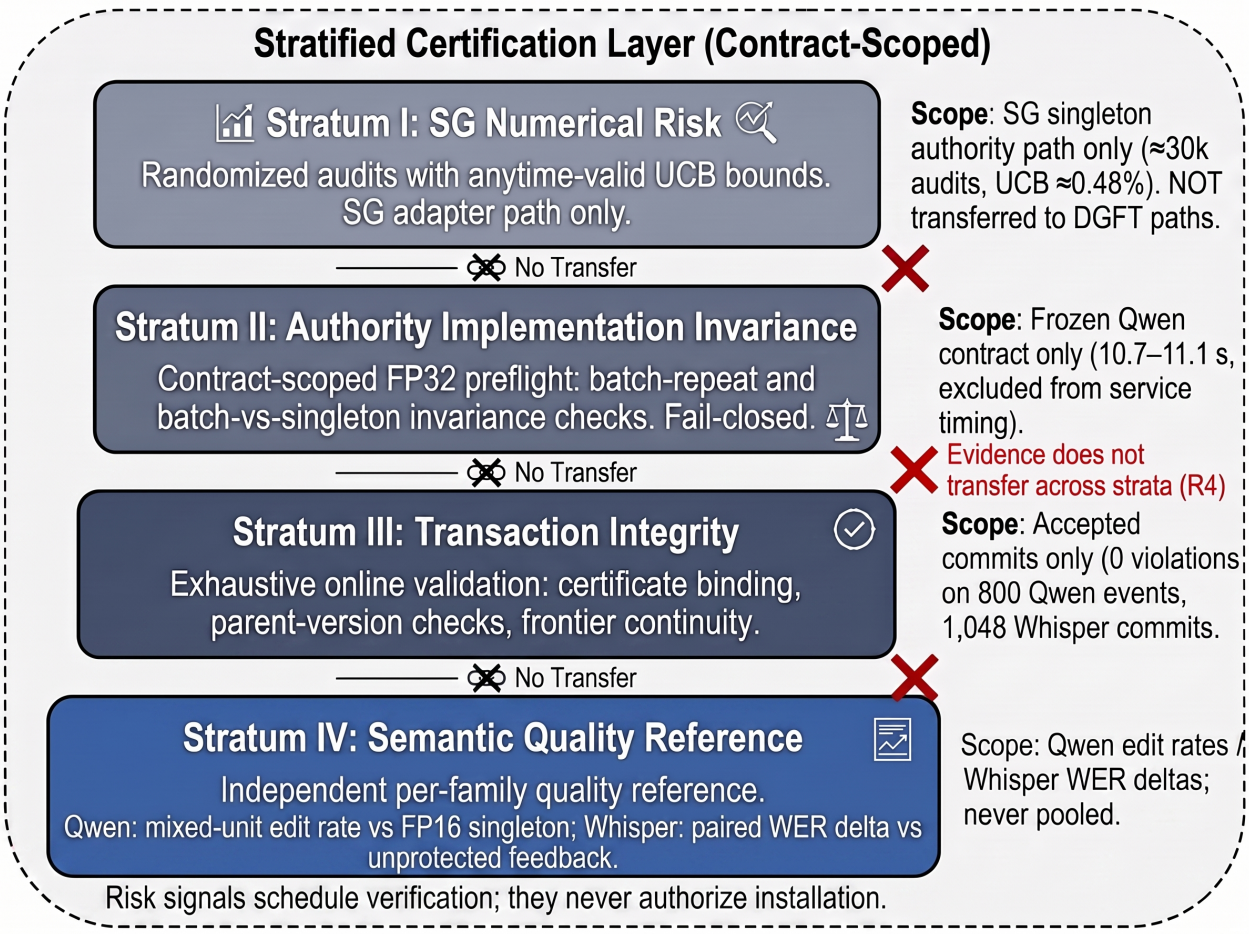


Fig. 3. Contract-scoped certification. Each stratum supports only its stated guarantee; evidence never transfers across strata. Qwen and Whisper quality estimands remain separate.

The primary deployment uses two RTX 3090 GPUs, one per plane, causal arrivals, and vectorized host transfer. The independent scale contract uses one RHEL 7.6 node with eight 32-GB V100 PCIe GPUs, PyTorch 2.4.1, CUDA 12.1, and Transformers eager execution. It runs up to four proposer and four authority replicas over a shared versioned commit queue—not distributed model parallelism. Audio is deterministically resampled to 16 kHz. Contract fingerprints, rejected realizations, raw records, and result hashes remain in the release; timing is never pooled across hardware contracts. Each reported run is bound to a code revision and contract digest so that protocol changes cannot be mixed with frozen measurements. A review artifact containing source code, frozen audited results, and analysis scripts accompanies this submission; a public archive will be released upon acceptance.

VI. Evaluation

A. Setup and Research Questions

We evaluate five checkpoints from two revision-generative ASR families: Qwen3-ASR 0.6B/1.7B and Whisper tiny/base/small.en. The main system runs use two RTX 3090 GPUs. The matched comparison uses the same 16 bilingual 30-s sessions (c16; 480 s of audio), whereas cross-family production results retain their original family-specific workloads and references. Scale-out is measured separately on one RHEL 7.6 node with eight 32-GB Tesla V100 PCIe GPUs and the Transformers-eager backend. Quality is evaluated against an independent canonical replay for Qwen and unprotected feedback for Whisper. We never pool results across hardware, families, authority paths, or quality estimands.

Metrics and timing. Audio xRT is completed audio seconds divided by service wall time. Wall speedup divides singleton-reference wall time by service wall time; GPU-normalized speedup additionally divides by the service-to-reference GPU-count ratio (two for the RTX 3090 cells). Energy is trapezoidally integrated from NVIDIA Management Library (NVML) power samples and normalized by completed audio seconds. Display latency measures ready-to-proposal completion, whereas state/segment-commit latency measures ready-to-authorized installation; the two clocks are never interchanged. Timed cells use immutable causal arrival schedules, while wall and energy meters record real execution. The matched Qwen cells use 2-s fine revisions and 6-s authority segments ($\rho = 3$). "16/16 pass" means that every Whisper production cell met its frozen thresholds: wall speedup $\geq 2.5\times$, GPU-normalized speedup $\geq 1.25\times$, energy reduction $\geq 20\%$, display P99 ≤ 1.9,s,

commit P99 $\leq$ 36,s, audio amplification $\leq 1.2\times$, and zero semantic violations.

The evaluation asks: *RQ1*, does the feedback edge causally propagate a local execution difference? *RQ2*, does DGFT preserve transaction integrity? *RQ3*, what performance–quality–energy frontier does the authority split expose, including against the nearest adapted baselines? *RQ4*, how does independent proposer/authority replication scale?

B. RQ1: Causal Propagation

We enumerate 21 legal Whisper state pairs and apply the same isolated batch-created divergence with the feedback edge present and removed, while holding future inputs, batch assignments, and arithmetic realization fixed. With feedback, 8/21 pairs propagate; without feedback, 0/21 do. Six pairs remain divergent into the measured tail, where the revision amplification factor (Definition II.6) reaches 153. This intervention identifies feedback as the propagation channel; it does not estimate a production prevalence.

C. RQ2: Transaction Integrity

We validate every accepted install in 800 Qwen events (480 SG and 320 DGFT) and 1,048 Whisper commits. Under the trust and binding assumptions of Section III, all completed sessions have monotone terminal frontiers, with zero observed transaction-install, output/state, stale-result, or replay violations. No stale install was attempted in these production traces; hence they validate accepted-history integrity rather than fault-injected stale rejection.

A deterministic state-machine fuzzer supplies the adversarial complement. It issues 250,000 attempts across 64 sessions and 24 invalid/replay classes, including cross-session and cross-epoch substitution, certificate tampering, stale parents, frontier gaps/overlaps, duplicates, and replays. The runtime accepts all 10,000 valid commits and rejects all 240,000 mutations, with no invariant violation. This is exhaustive validation over the evaluated traces and mutation suite, not a zero-failure-probability claim; it excludes Byzantine trusted components, GPU worker crashes, and wall-clock expiry.

To isolate transaction logic from accelerator execution, we benchmark the production commit primitive on the submission host using 31 randomized repeats of 20,000 valid calls (2,000-call warmup; 256-token prompt, 32-token display, and eight speculative descendants). The median full commit is 101.107 μs versus 19.077 μs for the same immutable-state materialization without checks, a paired incremental cost of 81.960 μs/commit; the full-commit P99 across repeat aggregates is 104.514 μs. This host-side protocol microbenchmark is not GPU or end-to-end speedup evidence; Supplementary Section S4 gives all variants and the existing online critical-section cross-check.

D. RQ3: Service Frontier, Quality, and Baselines

a) Matched authority configurations: Figure 4 compares complete configurations on the identical c16 workload: SG ($\rho = 1$, FP16 singleton authority), Coarse-DGFT ($\rho = 3$, FP32 batched authority), and Hybrid-DGFT ($\rho = 3$ plus selective

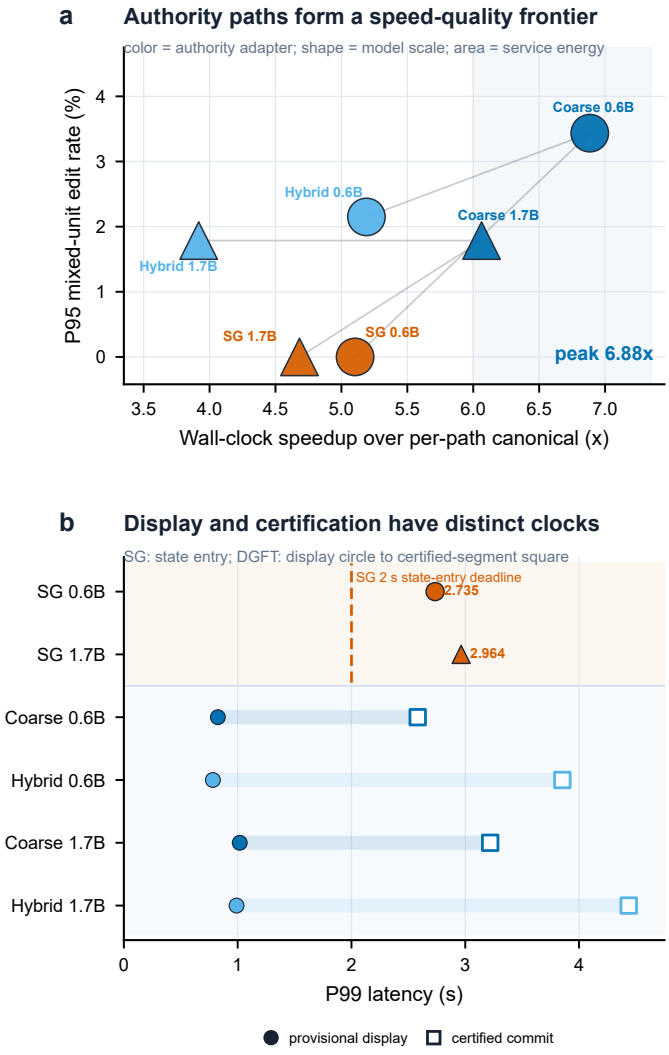


Fig. 4. Matched c16 authority configurations. Marker area denotes energy; display and certified-commit latencies remain distinct estimands.

replay). Authority precision, execution mode, and granularity change together, so this is not a precision-independent ablation of ρ.

Coarse reduces service wall time from 14.905 to 10.759 s (0.6B) and from 16.285 to 12.763 s (1.7B), giving direct throughput gains of $1.39\times$ and $1.28\times$ over SG. Its separately normalized headline speedups are $6.88\times$ /$6.06\times$ over each path's frozen all-singleton reference; these normalizations are not divided to obtain the matched ratios. SG's synchronized terminal burst yields certified state-entry P99 2.735/2.964 s and 2-s miss rates 3.33%/4.17%. Coarse/Hybrid instead expose display P99 $\leq$ 1.018 s and segment-commit P99 2.582–4.437 s. Coarse energy relative to SG changes from -4.4% on 0.6B to $+15.1\%$ on 1.7B; Hybrid costs $+16.8\%/+54.2\%$. The reversal confirms that the complete authority path, not granularity alone, determines the operating point.

b) Cross-family production and quality: On Qwen, Coarse-DGFT reaches $6.88\times$ /$6.06\times$ wall speedup, $3.44\times$ /$3.03\times$ GPU-normalized speedup, and 59.8%/51.0% energy reduction for 0.6B/1.7B. On Whisper, the same transaction

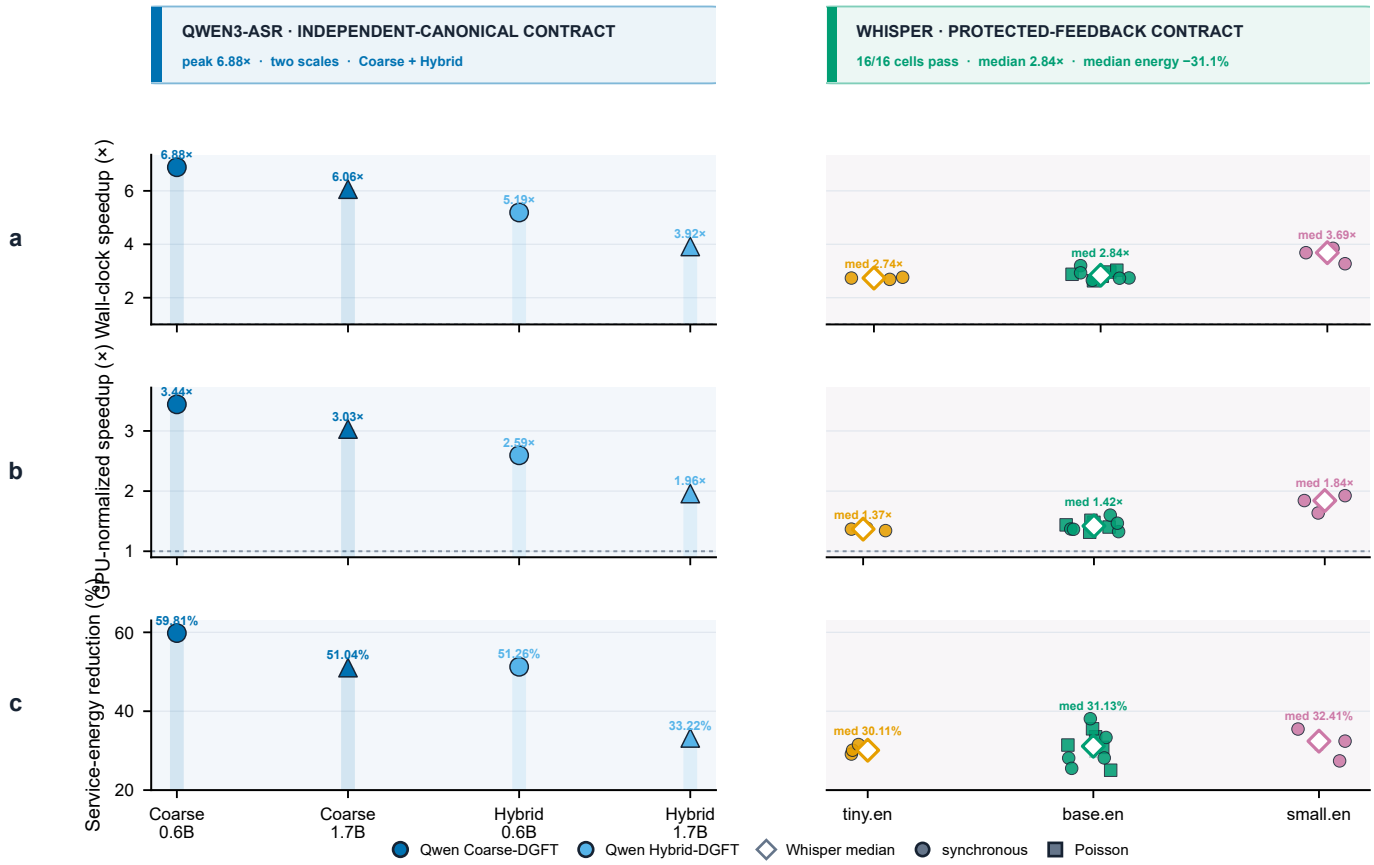


Fig. 5. Production performance under family-specific contracts. Qwen and Whisper use different baselines and are not pooled.

runtime passes all 16 production cells: median wall speedup is $2.841\times$ (range 2.633–3.849), GPU-normalized speedup is $1.421\times$, median energy reduction is 31.13%, and display P99 is at most 1.492 s. Figure 5 reports each family against its own baseline; the magnitudes are not compared across columns.

Qwen's mixed-unit edit rate against independent FP16 singleton replay has empirical P95 $\leq 3.43\%$ (worst session 5.41%; no catastrophic case). On deterministic 64-utterance holdouts from *LibriSpeech test-clean* and *AISHELL-1*, respectively, 0.6B/1.7B obtain English WER 2.15%/2.30% and Chinese CER 2.71%/2.16%, with 256/256 terminal outputs exactly matching singleton authority and partial-revision edit rates 24.7–31.8%. These are fixed holdouts, not full-corpus leaderboard results.

For Whisper, protection improves WER relative to unprotected feedback for tiny/base/small.en by $-12.23/-18.21/-25.27$ points on development sessions and $-8.26/-11.03/-13.76$ on the untouched cohort. All one-sided 95% bootstrap upper bounds remain below zero. The untouched cohort contains 40 long sessions per model, deterministically assembled from 214 previously unused source utterances; the long session is the bootstrap unit. These family-specific quality questions remain separate (Figure 6).

c) Nearest adapted baselines: Table II compares all methods on the independent V100 c16 workload; all 240 display events per cell meet the 2-s display SLO. Against an LLM-42-style revision adaptation that verifies every revision, Coarse-DGFT improves audio throughput by $6.91\times/5.64\times$ on 0.6B/1.7B. Against MarginGate-inspired scheduling, Coarse is 5.6% slower on 0.6B and 11.5% faster on 1.7B. The latter has no state-write authorization contract, so this comparison quantifies normal-load cost rather than equivalent safety.

On a separate overloaded c32 stress trace, MarginGate-inspired verification checks 3.02% of events but misses the known state-changing flip; an owner/version-only fence also accepts it. Always-verifying LLM-42-style replay removes the error at 1.565 events/s, whereas RevTrace verifies 5.05%, removes the error, and reaches 3.046 events/s ($1.95\times$). Because P99 is 6.107 s, this trace measures safety cost, not serving capacity. A legal execution-shape suite (batch size, slot, composition, and repeat) induces 1,021 proposal–canonical mismatches among 1,848 target-row evaluations; the frozen risk scheduler has zero observed false-safe events. Risk schedules verification but never certifies a write [13], [19].

E. RQ4: Replica Scaling and Capacity

The independent V100 study contains 84 cells spanning authority microprofiles, c16 baselines, c16–c128 replica allocations, and $\rho \in \{1, 3, 5\}$ attribution. We freeze the headline allocations and repeat 30 additional cells (three runs per model and configuration); ranges in Figure 7 are min–max. Repeated batched executions agree, but each model still has 62 batch–singleton request mismatches across precisions and batch sizes. FP32 batching is therefore a contract-scoped authority, not an unconditional singleton-equivalence certificate.

TABLE II
NORMAL-LOAD V100 C16 COMPARISON. EACH CELL HAS 16 BILINGUAL 30-S SESSIONS, 240 DISPLAY EVENTS, AND ZERO DISPLAY-SLO MISS OR OBSERVED INSTALL VIOLATION. P95 EDIT/EXACTNESS USE INDEPENDENT FP16 SINGLETON REPLAY. BASELINES ARE REVISION-LEVEL ADAPTATIONS.

Model	Mechanism	Verify/authority schedule	Audio xRT	Display P99	Commit P99	P95 edit	Exact sessions
Qwen3-ASR-0.6B	MarginGate-inspired	0.00%	40.501×	1.105 s	1.105 s	0.00%	16/16
	LLM-42-style revision	100%	5.535×	1.100 s	6.934 s	0.00%	16/16
	RevTrace SG	6.67%	29.474×	1.095 s	5.522 s	0.00%	16/16
	RevTrace Coarse	$\rho = 3$	38.223×	1.106 s	3.091 s	0.00%	16/16
	RevTrace Hybrid	$\rho = 3$ + replay	36.285×	1.086 s	3.334 s	0.00%	16/16
Qwen3-ASR-1.7B	MarginGate-inspired	3.33%	26.208×	1.433 s	5.061 s	0.00%	16/16
	LLM-42-style revision	100%	5.180×	1.393 s	7.837 s	0.00%	16/16
	RevTrace SG	10.00%	19.395×	1.440 s	7.891 s	0.00%	16/16
	RevTrace Coarse	$\rho = 3$	29.223×	1.609 s	4.492 s	3.18%	8/16
	RevTrace Hybrid	$\rho = 3$ + replay	27.811×	1.466 s	4.811 s	3.18%	8/16

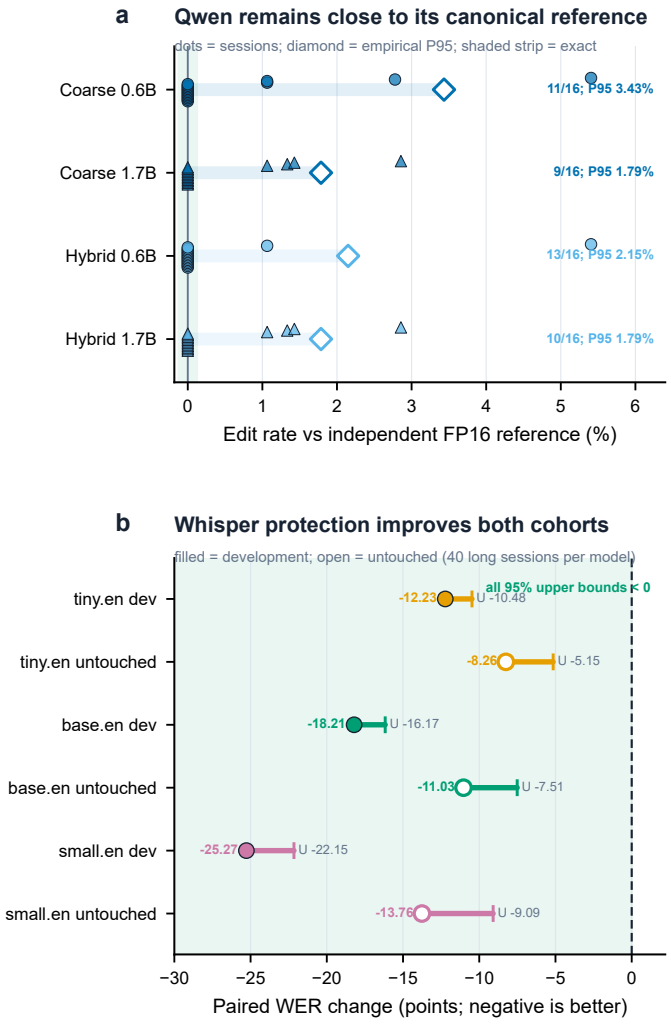


Fig. 6. Family-specific quality. Qwen measures edit distance to an independent canonical replay; Whisper measures paired WER change from unprotected feedback.

At c16, SG/Hybrid/Coarse reach $4.682/5.764/6.072\times$ wall speedup on 0.6B and $3.248/4.657/4.893\times$ on 1.7B. Balanced four-proposer plus four-authority replication at c64 sustains median $157.786 \times /111.722\times$ audio real-time throughput. Display P99 is 1.089/1.394 s with zero misses over 2,880 repeated events per model; certified-segment commit P99 is separately 3.080/4.890 s. At c128, display P99 rises to 2.079/2.730 s and commit P99 to 6.012/10.153 s, while throughput saturates. Thus c64 is the largest measured point satisfying the display SLO for both models, and c128 is retained as the failed boundary.

An authority-heavy 3+5 allocation at c64 lowers the authority/proposer service ratio but also lowers throughput to $93.788 \times /77.704\times$ and yields display P99 1.756/2.209 s; adding authority replicas does not repair the proposer bottleneck, and balanced 4+4 dominates. At $\rho = 1$, FP32 authority/proposer ratios 2.216/2.669 reduce throughput to $18.603 \times /12.294\times$; at $\rho = 5$, the ratios fall below one and throughput recovers to $40.630 \times /33.909\times$. These results support the two-stage cost model: balanced replication expands capacity, while coarse authority becomes economical only after sufficient amortization. They establish single-node, not multi-node, scaling.

VII. DISCUSSION AND BOUNDARIES

a) Authority semantics and evidence boundaries: FP32 batching is the canonical authority for the declared Coarse/Hybrid contract, not a bitwise repair of the FP16 proposer. DGFT requires an authority that emits a contract-bound projection and certificate; it does not require a particular precision. A deterministic low-precision kernel portfolio or canonical replay could replace FP32 only after a new contract epoch passes its own validation. The 10.7–11.1 s Qwen preflight is therefore reported once and excluded from service timing. Likewise, the SG adapter's 0.480% anytime-valid UCB over approximately 30,000 randomized audits does not transfer to Coarse/Hybrid, and Qwen edit distance to canonical replay is not pooled with Whisper WER change from unprotected feedback. Risk evidence may schedule verification, but only authority can certify a write.

b) Why the frontier is model dependent: The matched results support the complete cost decomposition rather than a granularity-only explanation. SG repeatedly invokes singleton authority and accumulates a synchronized terminal burst; Coarse amortizes authority work in a batch; Hybrid pays selective replay for a more conservative quality point. This mechanism explains the $1.39 \times /1.28\times$ Coarse gains and the cross-model energy reversal. SG therefore remains the exact same-granularity point, not a dominated baseline. Unlike token-level verified speculation [13], RevTrace authorizes a revision

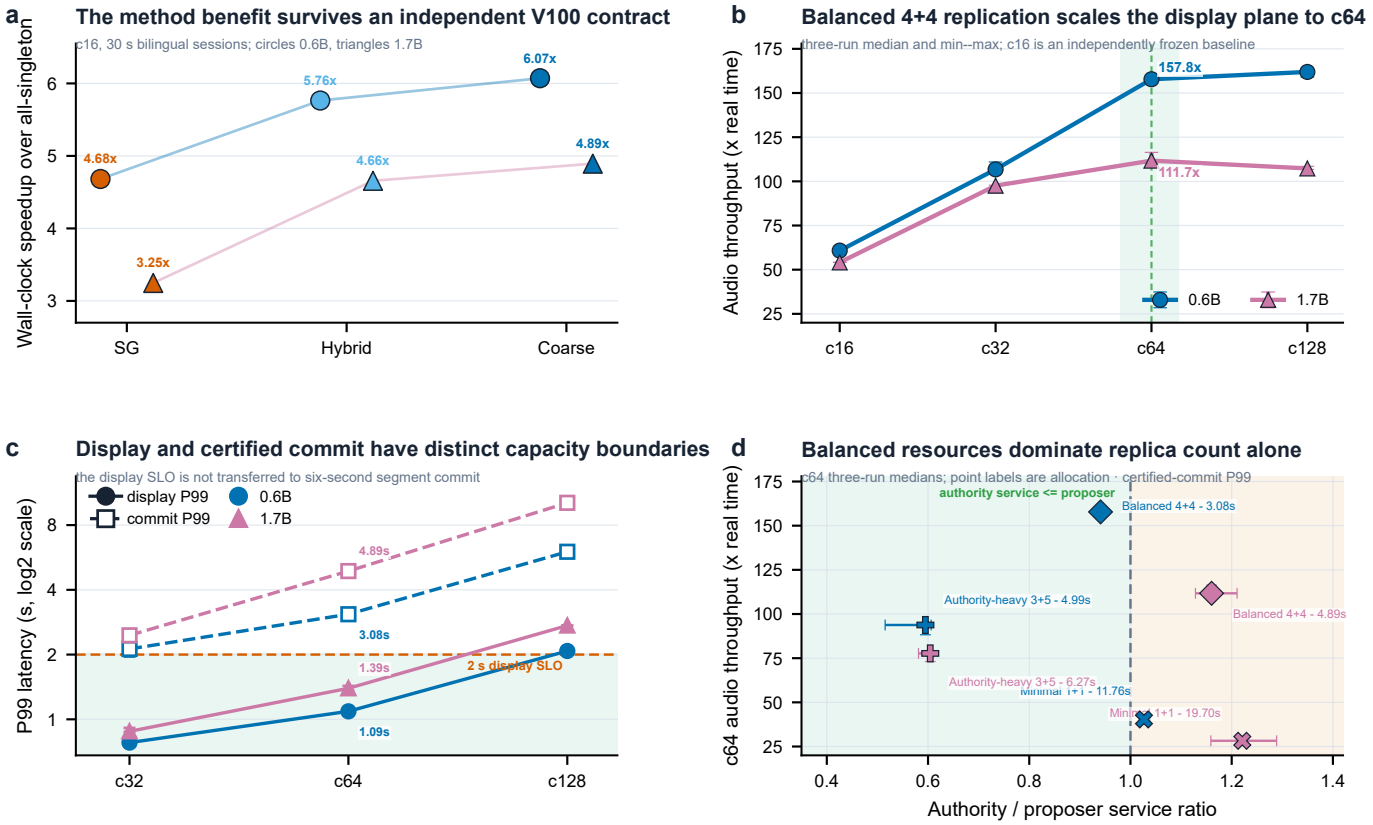


Fig. 7. Independent V100 replica scaling. Balanced 4+4 reaches the c64 display-SLO boundary; display and certified-commit latency are reported on distinct clocks. Points at c32–c128 show three-run medians and min–max ranges.

projection that can condition later rounds; unlike stateful cache reuse [20], [21], it governs the transition that creates reusable state. Display and certification also have distinct clocks: the c64 V100 point meets the 2-s display SLO while certified-segment commit P99 is 3.080/4.890 s. We state both rather than transferring the display label.

c) Applicability and limitations: DGFT targets pipelines where generated revisions are automatically fed back; append-only streaming designs without this edge need no such transaction boundary [22], [23]. The current proof assumes a non-Byzantine authority coordinator, immutable transaction inputs, collision-resistant digests, atomic certificate consumption, and parent-version comparison. Production traces contain no stale-install attempt, so stale rejection is guaranteed by the protocol and exercised by fuzzing, not estimated from production frequency. Evaluation covers two ASR architecture families and single-node accelerator systems; it establishes neither arbitrary generative-model generality nor multi-node throughput. A distributed realization can partition the session-local ledger and implement certificate consumption as a conditional write, but consensus latency, failover, and cross-node commit capacity remain future work.

VIII. Related Work

a) Streaming revision and feedback: Streaming ASR distinguishes partial-hypothesis stability from final recognition quality [24], [25]. Transducers revise hypotheses as audio arrives [26]; Whisper-based systems confirm prefixes after repeated agreement, and chunked variants detect truncation [1], [2], [27]. These policies control visible flicker. Self-refinement and agent loops also retain generated feedback [28], [29], but do not define when a numerically divergent revision may become an ancestor of future serving state. RevTrace separates display permission from this state-write authorization decision.

b) Verified and reproducible execution: Speculative decoding preserves a target distribution by verifying draft tokens [11], [12]; Medusa, EAGLE, and SpecInfer increase accepted work per verification step [30]–[32]. LLM-42 instead combines a fast nondeterministic path with fixed-path verification and rollback, while MarginGate uses token margins to schedule that verification [13], [19]. RevTrace verifies a different object: a revision projection that may condition all later rounds, bound to parent provenance and an input frontier. Numerical nonidentity from reductions, kernel selection, and low precision is well documented [10], [33]; reproducible reductions can impose implementation or performance costs [34], [35]. Rather than require a bitwise-deterministic proposer, RevTrace names a canonical authority and gates state installation. Anytime-valid statistics [17], [18], [36] monitor one declared SG audit process; they do not certify individual transitions.

c) Serving and materialized state: Contemporary serving systems improve throughput through paged KV management, iteration scheduling, structured runtimes, chunked prefill, phase disaggregation, and migration [4]–[9]. Stateful systems additionally reuse, transport, or schedule materialized KV and

application state [20], [21], [37]–[39]. These mechanisms govern execution, residency, and reuse after state exists. RevTrace is complementary: it controls whether a batch-created model projection is authorized to enter that reusable state in the first place.

d) Concurrency-control basis and remaining gap: DGFT adapts validation and version fencing from optimistic and multiversion concurrency control [14], [15]. Its invalidation model is related to Time Warp rollback, while provisional external effects recall the rollback/compensation distinction in Sagas [40], [41]. The guarantee is narrower than general linearizability [16]: it proves one fork-free committed revision history under certificate and parent-version rules, not a real-time specification for all client operations. The adjacent lines thus protect display stability, token acceptance, numerical execution, or already materialized state. This leaves an unresolved systems gap: to our knowledge, they do not jointly bind a generated feedback projection to canonical authority, its parent version, and its input frontier before installation. RevTrace targets precisely this intersection without claiming to replace those numerical, caching, or scheduling mechanisms.

IX. Conclusion

Revision-generative ASR turns an output difference into a speculative write to future model state. RevTrace addresses this failure mode with DGFT: proposal and display remain fast, while only a certificate bound to canonical authority, parent version, and input frontier can install feedback state. The protocol excludes uncertified ancestry and forked committed histories under its stated trust assumptions. Causal intervention confirms the feedback channel (8/21 legal pairs propagate versus 0/21 without feedback), and the evaluated traces show no installation violation across 800 Qwen events and 1,048 Whisper commits; a 250,000-attempt fuzzer rejects every invalid/replayed mutation. Coarse authority improves matched throughput by $1.39\times/1.28\times$, while family-specific production contracts reach up to $6.88\times$ wall speedup. Independent $8\times$V100 results identify c64 as the largest measured display-SLO-valid point, sustaining $157.8\times/111.7\times$ audio real-time throughput. These results show that state-write authorization can be both a formal runtime contract and a practical parallel-serving primitive. Current evidence is limited to two ASR families and single-node systems; evaluating a session-partitioned ledger under multi-node consensus and additional feedback-driven model classes is future work.

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